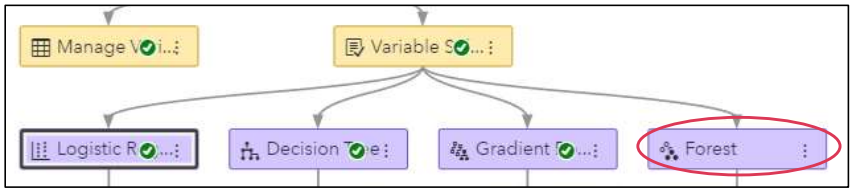
Modeling a Binary Target with a Forest

In this demonstration, you add a Forest node to the Lesson 3 pipeline. You build a default forest model, change some of the settings, and compare the model to the other models in the pipeline.

1. In the Lesson 3 pipeline, right-click the **Variable Selection** node and select **Add child node ⇨ Supervised Learning ⇨ Forest**.



2. Keep all properties for the forest at their defaults. Run the **Forest** node.
3. Open the results for the node.
4. Click the **Assessment** tab.

Target ...	Data Role	Partitio...	Formatt...	Sum of ...	Averag...
churn	TRAIN	1	1	39,590	0.0506
churn	VALIDATE	0	0	16,967	0.0591

The Fit Statistics table shows an average squared error of 0.0591 on the VALIDATE partition.

This performance is again pretty good, better than the decision tree tuned in the earlier demonstration (but not quite as good as for the gradient boosting models). But again, try to improve the forest performance by changing some of the default settings.

5. Close the Results window.
6. The random forest is an ensemble of many decision trees, so it stands to reason that the number of trees will have a significant effect on the resulting model accuracy. Reduce **Number of trees** from 100 to **50**.

   In general, the more trees you use tends to improve the performance of the forest. However, the improvement decreases as the number of trees increases. In other words, at a certain point, the benefit in prediction performance from learning more trees will be lower than the cost in computation time for learning these additional trees.

   If you do not have any concern regarding the computation times, the more trees you have, the better (reliable) estimates you get from out-of-bag predictions. You can use OOB error rate to determine the number of trees.

7. Under the Tree-splitting Options properties, change **Class Target Criterion** from Information gain ratio to **Entropy**.

   Information gain ratio is a measure of purity. Entropy on the other hand is a measure of impurity (the opposite). Entropy controls how a decision tree decides to split the data. It affects how a decision tree draws its boundaries.

8. Decrease **Maximum depth** from 20 to **12**.

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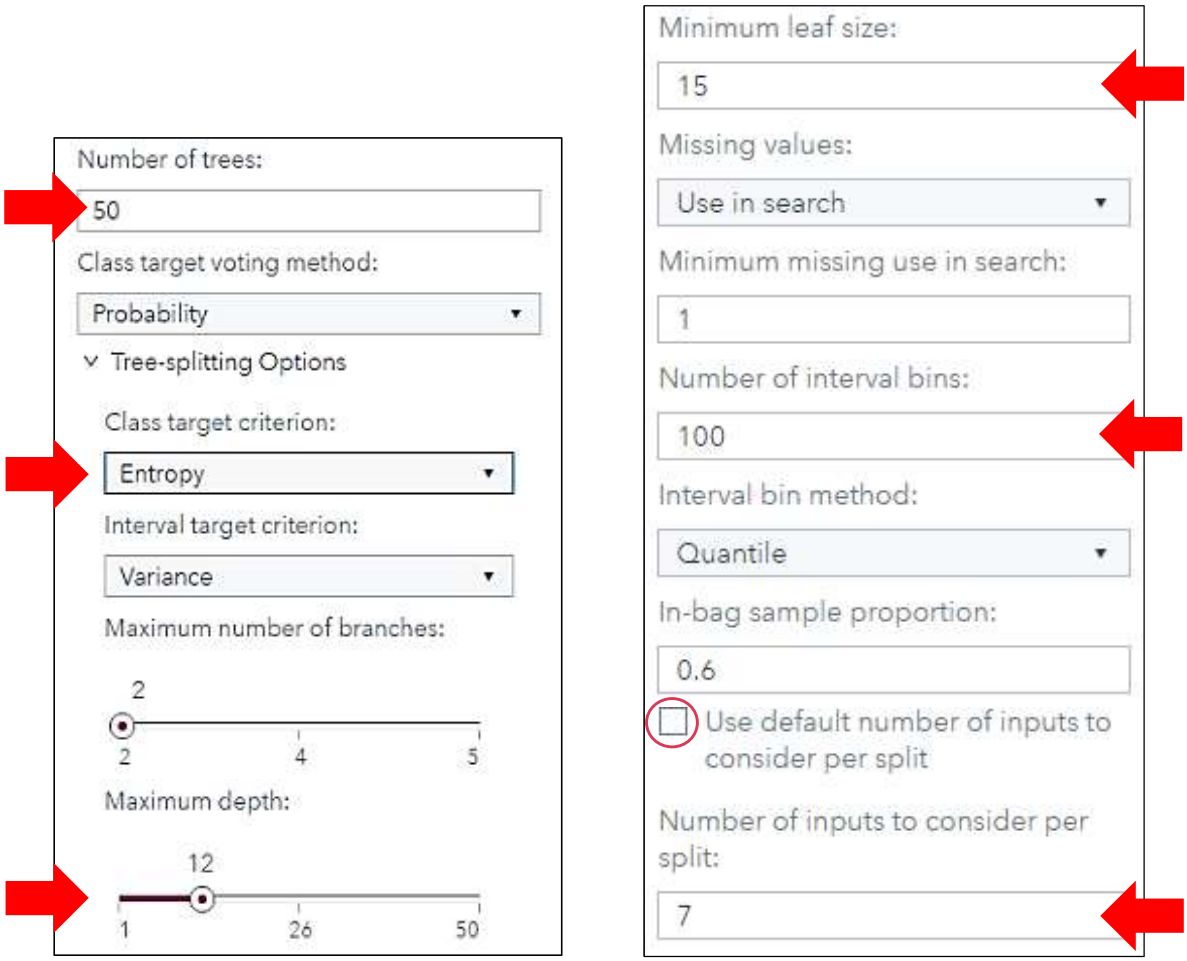
9. Increase **Minimum leaf size** from 5 to **15**.

   The decision trees in a random forest are overtrained by letting them grow to a large depth (default maximum depth of 20) and small leaf size (default smallest number of observations per node of 5). The idea behind this approach is that averaging the predicted probabilities of a large number of overtrained trees is more robust than using a single fine-tuned decision tree.

10. Increase **Number of interval bins** from 50 to **100**.

11. Clear the box for the option **Use default number of inputs to consider per split**, which by default is the square root of the number of available inputs. Set this parameter to **7** (half of inputs).

    This is another key aspect of random forests: how many variables to consider for splitting each node (that is, a random subset of all variables, as opposed to considering all variables). This might reduce the bias toward the most influential variables and allows for a more generalizable model. You can manually change this option over numerous training runs.



12. Run the **Forest** node.

13. Open the results for the node.

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14. Click the **Assessment** tab.

Target ...	Data Role	Partitio...	Formatt...	Sum of ...	Averag...
churn	TRAIN	1	1	39,590	0.0520
churn	VALIDATE	0	0	16,967	0.0593

The average squared error for the tuned forest model is 0.0593 on the VALIDATE partition. This fit statistic is a little bit worse than the first model by using the default settings.

15. Close the Results window.
16. Run the entire pipeline and view the results of model comparison.

Champion	Name	Algorithm Name	KS (Youden)	Misclassification Rate
✓	Gradient Boosting	Gradient Boosting	0.5836	0.0648
	Forest	Forest	0.5734	0.0700
	Decision Tree	Decision Tree	0.5480	0.0711
	Logistic Regression	Logistic Regression	0.5338	0.0815

The gradient boosting model is now the champion model of the pipeline, based on default KS. The forest is a close second-place model.

17. Close the Results window.

End of Demonstration

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